Table 1: Probability-based performance for methods with a defined estimate of the population positive-class probability. Brier score and log loss are computed from final train+validation refit probabilities; lower values indicate better probability predictions. Ranking-only scores without a population-probability interpretation are not assigned Brier or log-loss values. Aggregation follows the benchmark weighting scheme described in the text.

Panel A: Original controlled simulations

Method	Brier OS	Brier TS	Log loss OS	Log loss TS
Logistic probability correction	0.1413	0.1414	0.4587	0.4687
uPU logistic	0.1411	0.1419	0.4427	0.4437
nnPU logistic	0.1411	0.1413	0.4427	0.4436
Corrected AdaBoost	0.3432	0.3883	1.9047	2.5072
Corrected GBM	0.1432	0.1458	0.4723	0.4935
Corrected XGBoost	0.1411	0.1420	0.4492	0.4601
Corrected LightGBM	0.1436	0.1457	0.4661	0.4868
Corrected CatBoost	0.1411	0.1419	0.4470	0.4552
Supervised logistic regression (supervised reference)	0.1389	0.1381	0.4368	0.4339
Supervised CatBoost (supervised reference)	0.1376	0.1368	0.4332	0.4303

Panel B: Nonlinear simulation extension

Method	Brier OS	Brier TS	Log loss OS	Log loss TS
Logistic probability correction	0.1626	0.1639	0.5451	0.5698
uPU logistic	0.1627	0.1726	0.5062	0.5311
nnPU logistic	0.1627	0.1643	0.5062	0.5182
Corrected AdaBoost	0.2633	0.3063	0.7772	1.1154
Corrected GBM	0.1425	0.1445	0.4814	0.5014
Corrected XGBoost	0.1373	0.1389	0.4435	0.4539
Corrected LightGBM	0.1393	0.1411	0.4560	0.4684
Corrected CatBoost	0.1338	0.1349	0.4288	0.4327
Supervised logistic regression (supervised reference)	0.1571	0.1573	0.4919	0.4922
Supervised CatBoost (supervised reference)	0.1258	0.1258	0.4030	0.4029

Panel C: Real-data experiments

Method	Brier OS	Brier TS	Log loss OS	Log loss TS
Logistic probability correction	0.1043	0.1053	0.3809	0.3814
uPU logistic	0.1009	0.1030	0.4425	0.4511
nnPU logistic	0.0968	0.0978	0.3257	0.3302
Corrected AdaBoost	0.5594	0.5812	7.2552	7.7422
Corrected GBM	0.1139	0.1159	0.4114	0.4156
Corrected XGBoost	0.1015	0.1012	0.3659	0.3676
Corrected LightGBM	0.1266	0.1354	0.5059	0.5229
Corrected CatBoost	0.1036	0.1032	0.3552	0.3538
Supervised logistic regression (supervised reference)	0.0690	0.0691	0.2320	0.2325
Supervised CatBoost (supervised reference)	0.0549	0.0552	0.1834	0.1842

Note. OS: one-sample; TS: two-sample. Panel A: 100 repetitions per setting, settings averaged within ten families, then equal family weights. Panel B: 100 repetitions per DGP, then equal weights for six DGPs. Panel C: 20 splits per dataset/design/budget, then equal weights for two budgets and eight datasets. All ten eligible methods have complete coverage in all panels. The final two methods in each panel are supervised references, not practical PU competitors. Values describe probability prediction accuracy, not calibration alone.